

Synthesizing Strategic Insight: Evaluating Fine-Tuning and Agentic RAG for Professional-Grade Sports Commentary

Kendrix Henderson, Navnith Bharadwaj

Course CSCI 5922, University of Colorado Boulder

Abstract. The abstract should summarize the contents of the paper. The length can be around 70 to 150 words.

Keywords: We would like to encourage you to list your keywords within the abstract section

1 Introduction

Paragraph 1: Explain the motivation for your work; e.g., Why anyone should care? What are the desired benefits?

Paragraph 2: Explain why existing solutions are inadequate for the motivated problem; e.g., Is there a gap in what exists? Is there a weakness in existing approaches?

Paragraph 3: Explain what you are proposing, what is new about your idea, and why you believe this solution will be better than previous solutions; e.g., Are you asking a new question, establishing a new methodology to solve a problem, building a new software tool, or offering greater understanding about existing methods/tools?

2 Related Work

Automatic Sports Commentary Generation

- [1]
- [2]
- [3]

The research regarding intelligent autonomous sports commentary is extremely sparse as often regarded within each sources related works. Formula one is not short of research, however within this domain the work is within infancy. Our work which involves applying such commentary to Formula 1 is a challenge only made possible via similar works done within baseball and basketball. The work undergone by Lee on [2] provides an insight on how we can make our commentary more insightful from an automated narrative.

DPO vs. RLHF

- [4]
- [5]
- [6]

A prime motivation within commentary been to provide listeners with insightful, most importantly entertaining dialouge. In previous contexts regarding large language models and the broader scope of natural language tasks, we have seen advancements in tailoring responses to match human preference. A breakthrough within this realm has come from the explorations undergone by Ouyang [5]. Although the benefits from introducing RLHF, can greatly improve the commentary generated an approach found by Rafailov [4] would reap the benefits of such implemntation without requiring as much overhead. Both are notable potentail additions upon our proposed methodology, however with the added cost of compute or dataset additions we are unable to implement such without risk of straying from the affordance of being consumer feasible and achievable within a month.

Proper Dataset Creation

- [7]

In contrast to exisiting datasets and achievments from similair works such as Zhang [7], Formula 1 constantly makes regulation overhauls on average every 6-8 years. This creates an issue in dataset creation as we are unable to use a significantly large set of previous years telemetry data. Additionally we were unable to ethically source visual data which resulted in an absence of visual media within our dataset. In the creation of our dataset we used strategies learned from Zhang, to create a suitable dataset based upon the most up to date reglations within our targeted sport. We relied on historical telemetry data from OpenF1 and publicly available ground truth commentary.

3 Methodology

Methods - describe the implementation of your proposed idea (e.g., description of your model, system, or how survey papers will be collected) so that:

- A reader could reproduce your set-up
- A reader could understand why you made your design decisions

4 Experiments

Experiments - describe 1-2 experiments or analyses you plan to conduct in order to demonstrate/validate the target contribution(s) of your work. Your description should be detailed enough so that a reader could reproduce it. Your description should include the following for each experiment:

- Main purpose: 1-3 sentence high level explanation
- Evaluation Metric(s): which ones will you use and why?

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5 Conclusions

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The paper ends with a conclusion.

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References

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1. Kim, B.J., Choi, Y.S.: Automatic baseball commentary generation using deep learning. In: Proceedings of the 35th Annual ACM Symposium on Applied Computing. (2020) 1056–1065

2. Greg, L., Vadim, B., Elliot, L.: Sports commentary recommendation system (scores): Machine learning for automated narrative. Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment **8** (2012) 32–37

3. Jung, S., Kim, H., Park, H., Choi, A.: Integrated ai system for real-time sports broadcasting: Player behavior, game event recognition, and generative ai commentary in basketball games. Applied Sciences **15**(3) (2025)

4. Rafailov, R., Sharma, A., Mitchell, E., Ermon, S., Manning, C.D., Finn, C.: Direct preference optimization: Your language model is secretly a reward model (2024)

5. Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C.L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askill, A., Welinder, P., Christiano, P., Leike, J., Lowe, R.: Training language models to follow instructions with human feedback (2022)

6. Pareja, A., Nayak, N.S., Wang, H., Killamsetty, K., Sudalairaj, S., Zhao, W., Han, S., Bhandwaldar, A., Xu, G., Xu, K., Han, L., Inglis, L., Srivastava, A.: Unveiling the secret recipe: A guide for supervised fine-tuning small llms (2024)

7. Zhang, B., Gao, J., Yuan, Y.: A descriptive basketball highlight dataset for automatic commentary generation. In: Proceedings of the 32nd ACM International Conference on Multimedia. MM '24, New York, NY, USA, Association for Computing Machinery (2024) 10316–10325